

Implementation status summary (v7)	
♥ Fulfilled	27
▲ Partial	6
✖ Not fulfilled	6
● Hardware / out of scope	9
! N/A	1

Row highlight: pale yellow = new or significantly changed vs. PZEMESP32 v4
cos phi note: $\phi_{\text{I apparent}} = \arccos(\text{I} / \sqrt{\text{V} \times \text{I}})$. Accurate for resistive loads; approximate for SMP3/phase-cut loads (mixers, blenders). Column renamed in CSV header to avoid confusion with PZEM direct hardware measurement